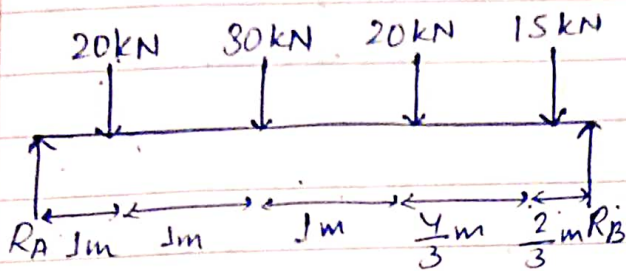


Find out the support reactions (R_A, R_B) of the spring supported beam shown above.

- ① Simple / Roller support $\rightarrow$ 1 reaction force.
- ② Hinged support $\rightarrow$ 2 reaction force.
- ③ Fixed support $\rightarrow$ 2 reaction force + 1 reaction moment.



$$\sum F_y = 0$$

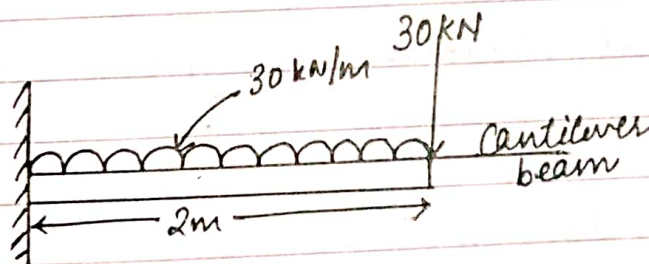
$$R_A + R_B = 20 + 30 + 20 + 15 = 85 \text{ kN} \quad \text{--- (1)}$$

$$\sum M_A = 0$$

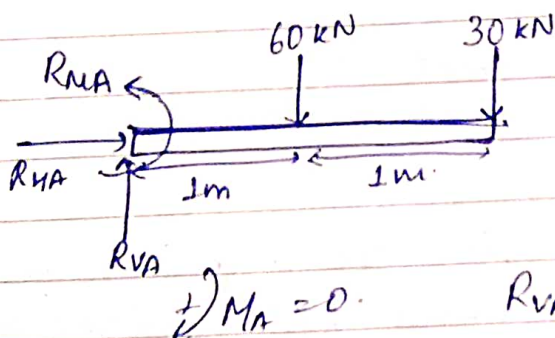
$$20 \times 1 + 30 \times 2 + 20 \times 3 + 15 \times 5 = R_B \times 5 \quad \text{--- (2)}$$

$$R_A = 44 \text{ kN} \quad R_B = 41 \text{ kN}$$

Ques



Find the support reaction and moment at fixed end



$$R_{MA} = 60 \times 1 + 30 \times 2 = 60 + 60 = 120 \text{ kNm}$$

$$\sum F_x = 0 \quad R_{HA} = 0$$

$$\sum F_y = 0$$

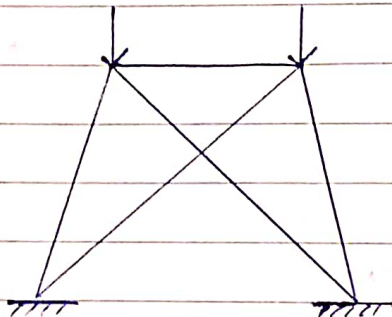
$$R_{VA} - 60 - 30 = 0 \quad R_{VA} = 90 \text{ kN}$$

TRUSS / PIN- JOINTED FRAME

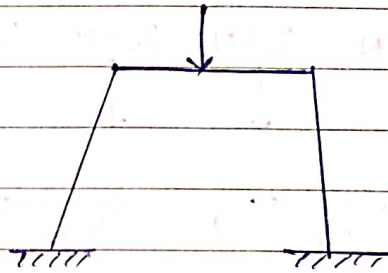
Truss is essentially a rigid structure made up of collection of straight members joined with each other through pin joints.

Assumption for a truss:-

- ① Members are connected by pin joints, which are free to rotate.
- ② Members are subjected to only axial loads, so there will be no bending moments.
- ③ Negligible weight of members.



TRUSS



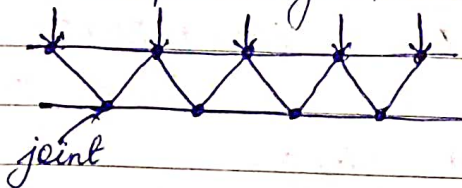
FRAME

Loads are acting on pin joints only. So, they can carry axial loads only.

No joints are there and bending moments are present in frames.

- * Triangular truss do not deform. Can form large truss. Most stable.

Simplest perfect structure and larger structure can be build up using triangular truss.



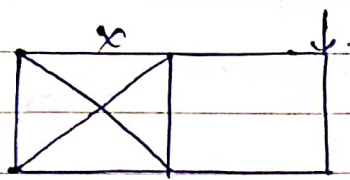
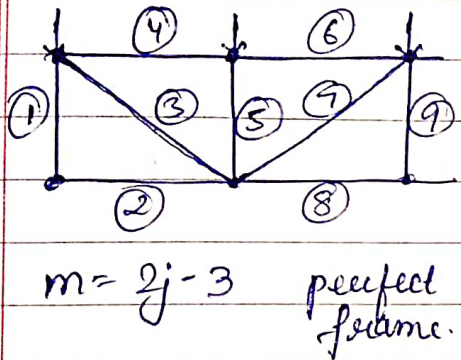
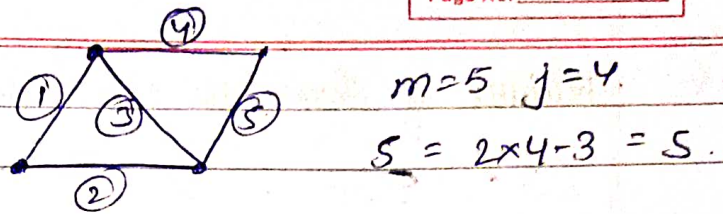
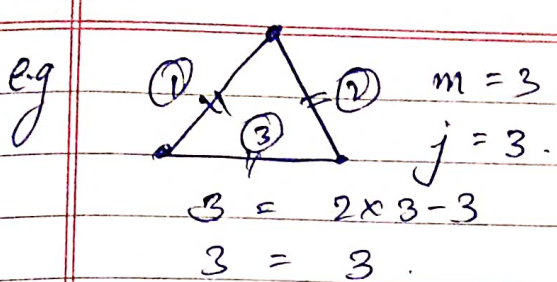
Plane truss.

- ① Perfect Frame → If a frame has got sufficient number of members, so that it can resist the applied loads, called perfect frame. i.e. it will not undergo any deformation.

necessary condition.

$$m = 2j - 3$$

m = number of members
 j = no. of joints.

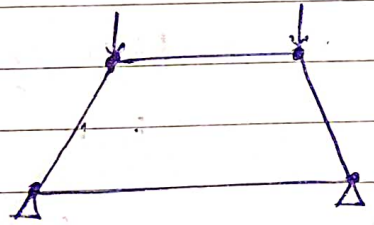


$m = 2j - 3$
 not a perfect frame although satisfies

* The sufficient condition for a perfect frame is that it satisfies necessary condition ($m = 2j - 3$) as well as that undergo any deformation under applied loads.

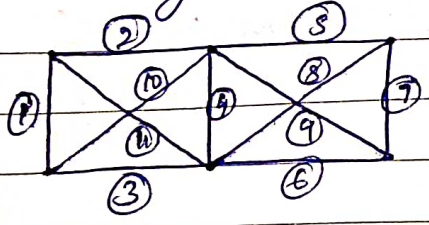
② Deficient Frame:- If number of members in a frame is less than what required to be a perfect frame, is called deficient frame. It will deform under applied loads.

$m < 2j - 3$



$m = 4$ $j = 4$
 $4 < 2 \times 4 - 3$
 $4 < 5$

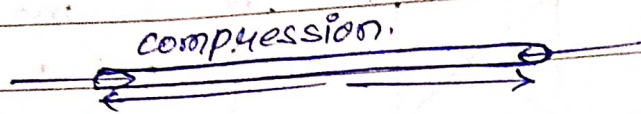
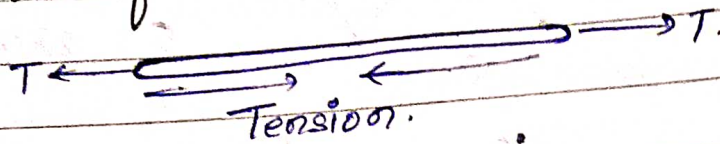
③ Redundant Frame:- If number of members in a frame is greater than what required to be a perfect frame, is called redundant frame. Each extra member will add one degree of indeterminacy so, it is called statically indeterminate frame.



$m = 11$ $j = 6$
 $11 > 2 \times 6 - 3$
 $11 > 9$

2 extra members, hence, 2 degree redundant frame.

Nature of Forces in Member:-



Method of Analysis:-

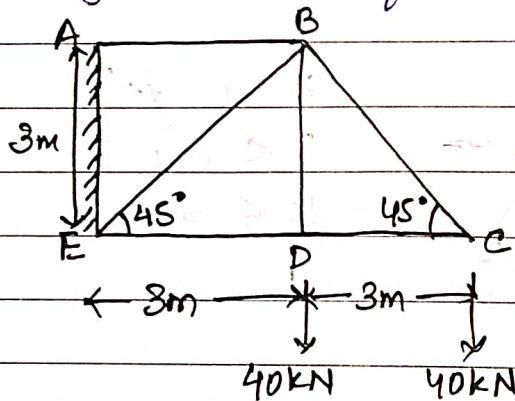
① Method of joint

② Method of Section.

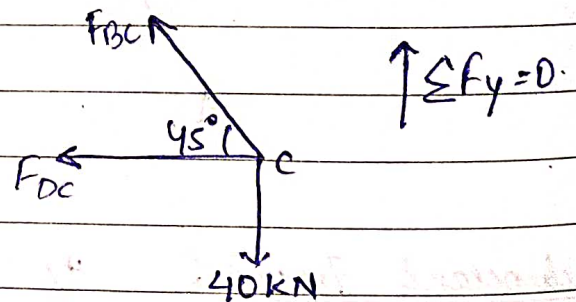
Joint Method: Look for a joint where there are only two unknown are present. If such a joint is not there then find the support reaction. then no. of unknown reduce to two at joint.

Apply the conditions of equilibrium ($\sum F_x = 0$ $\sum F_y = 0$) at the joint and find the forces in unknown member.

Ques



FBD of joint C.



Find the forces in all members of truss shown. tabulate the results.

$$F_{BC} \sin 45^\circ - 40 = 0$$

$$F_{BC} = 56.57 \text{ kN (T)}$$

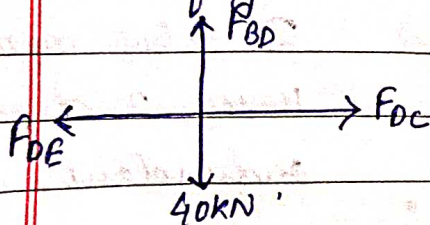
$$\rightarrow \sum F_x = 0$$

$$-F_{BC} \cos 45^\circ - F_{DC} = 0$$

$$F_{DC} = -F_{BC} \cos 45^\circ$$

$$F_{DC} = -40 \text{ kN (C)}$$

FBD of joint D.



$$\rightarrow \sum F_x = 0$$

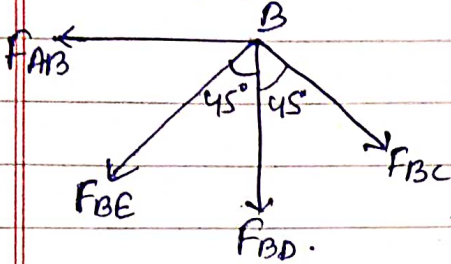
$$\therefore F_{DE} = F_{DC}$$

$$\therefore F_{BE} = -40 \text{ kN (C)}$$

$$F_{BD} - 40 = 0$$

$$F_{BD} = 40 \text{ kN (T)}$$

FBD of joint B.



$$F_{AB} + F_{BE} \sin 45^\circ = F_{BC} \sin 45^\circ$$

$$F_{AB} + \frac{F_{BE}}{\sqrt{2}} = \frac{F_{BC}}{\sqrt{2}}$$

$$F_{AB} + \frac{(-113.13)}{\sqrt{2}} = \frac{56.57}{\sqrt{2}}$$

$$F_{AB} - 79.99 = 40.00$$

$$F_{AB} = 120 \text{ kN (T)}$$

$$+\uparrow \sum F_y = 0$$

$$-(F_{BE} \cos 45^\circ + F_{BC} \cos 45^\circ + F_{BD}) = 0$$

$$40 + F_{BD} + F_{BE} \cos 45^\circ = 0$$

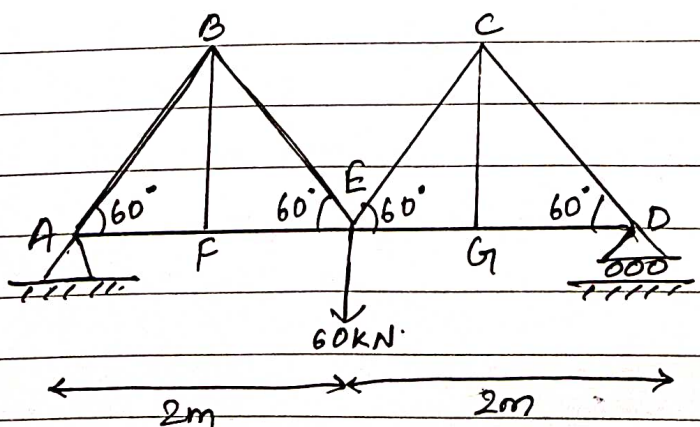
$$40 + 40 = -F_{BE} \times \frac{1}{\sqrt{2}}$$

$$F_{BE} = -80 \times \sqrt{2}$$

$$= -113.13 \text{ N (C)}$$

Sl. No	Force in member	Magnitude	Nature
1.	F_{BC}	56.57 kN	T
2.	F_{DC}, F_{ED}	40 kN	C
3.	F_{BD}	40 kN	T
4.	F_{BE}	113.13 kN	C
5.	F_{AB}	120 kN	T

Ques



Find the Force in all the members of truss and Tabulate the results.